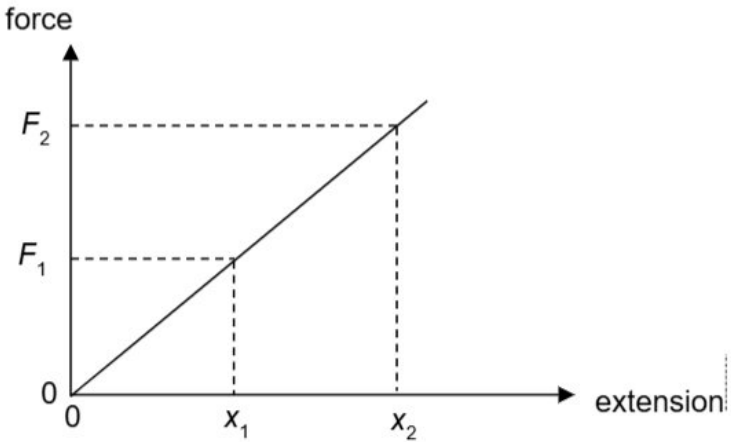
3 The force-extension graph of a spring is as shown.



What is the change in elastic potential energy when the extension of the spring is increased from x_1 to x_2?

A $F_2x_2 - F_1x_1$

B $\frac{1}{2}(F_2x_2 - F_1x_1)$

C $F_2x_2^{\,2} - F_1x_1^{\,2}$

D $\frac{1}{2}\left(F_2x_2^{\,2} - F_1x_1^{\,2}\right)$